

Mohammed El Kassoiri

Junior Data Scientist | Machine Learning & AI Engineer

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PROFESSIONAL SUMMARY

Junior Data Scientist and Machine Learning Engineer with hands-on experience in deep learning, computer vision, geospatial AI, and NLP. Proven ability to design end-to-end AI pipelines from raw satellite data ingestion to production API deployment with a strong track record in semantic segmentation, RAG systems, and GPU-accelerated workflows. Seeking to apply expertise in a collaborative, research-driven environment to solve complex scientific and societal challenges.

EDUCATION

Diplôme d'Ingénieur en Science des Données, Big Data et Intelligence Artificielle (Engineering Degree) <i>École Nationale Supérieure de l'Intelligence Artificielle et des Sciences des Données (ENSIASD), Morocco</i>	2023 - 2026
Relevant coursework: Machine Learning, Deep Learning, Big Data, Computer Vision, NLP, Distributed Computing, Statistics	
CPGE - Physics & Engineering Sciences (PSI) <i>Lycée Ibn Timiya, Marrakech, Morocco</i>	2021 - 2023

PROFESSIONAL EXPERIENCE

Intern - Automatic Agricultural Plot Delineation (PFE) <i>Pôle Digital de l'Agriculture, de la Forêt et Observatoire de la Sécheresse, Morocco</i>	Feb 2026 - Present (Expected Jun 2026)
- Designed an end-to-end satellite data pipeline, shapefile ingestion, QGIS processing, rasterisation, tiling, RGB + mask dataset generation, reducing manual preprocessing time by ~60%. - Developed an improved DeepLabV3+ segmentation model integrating RGB + Edge bands (4-channel input), achieving field boundary delineation on multi-source satellite data. - Integrated Sentinel-2 imagery via Google Earth Engine (GEE) to enable multi-scale geographic generalisation across diverse agricultural regions. Technologies: Python, TensorFlow, OpenCV, QGIS, Google Earth Engine, Semantic Segmentation, Remote Sensing, GIS.	
Data Science & Machine Learning Intern <i>XAI Morocco</i>	Aug 2025 - Oct 2025
- Contributed to the X-EdTech AI platform, building three production-ready modules: X-Forms (AI form generation), X-Quiz (automated question creation), and X-Table (data visualisation and analysis). Technologies: Python, FastAPI, Scikit-learn, NLP, Data Analysis, Backend Development.	

ACADEMIC PROJECTS

Legal AI - MY LAW: Moroccan Legal RAG System <i>Academic Project - ENSIASD</i>	2025
- Built a multilingual (Arabic/French) retrieval-augmented generation (RAG) system for intelligent question-answering over Moroccan legal texts, deployed via FastAPI and Gradio. - Implemented optimized semantic search using FAISS vector indexing and SentenceTransformers, achieving sub-second retrieval latency across 10,000+ legal documents and ensured legal accuracy by enforcing strict citation of law articles in every generated response, reducing hallucination risk. - Technologies: NLP, RAG, FAISS, SentenceTransformers, FastAPI, Gradio, LLMs, Multilingual AI.	
Water Stress Mapping - Taroudant Region <i>Academic Project - ENSIASD</i>	2025
- Designed a semantic segmentation system for agricultural water stress mapping, fusing 16 spectral bands (optical + SAR radar + geospatial indices). - Implemented an Attention U-Net model achieving 0.92 overall accuracy and 0.50 Dice coefficient on imbalanced satellite imagery, outperforming standard U-Net baselines, using a weak supervision pipeline on Google Earth Engine, eliminating the need for fully annotated ground truth labels. Technologies: Remote Sensing, Deep Learning, Attention U-Net, GEE, Weak Supervision, PyTorch.	
Robot Arm Object Sorting System <i>End-of-Year Project - ENSIASD</i>	2024
- Built a real-time object detection and automated colour-based sorting system integrated with a physical robot arm. - Achieved high-precision colour recognition using OpenCV with <50ms latency for real-time sorting decisions. Technologies: Computer Vision, OpenCV, Real-time Systems, Robotics Integration.	

TECHNICAL SKILLS

Programming And MLOps	Python, SQL, NumPy, Pandas, SciPy, Matplotlib, Seaborn, FastAPI, Git, Linux, Jupyter, Docker (basic)
ML / Deep Learning	Scikit-learn, PyTorch, TensorFlow, Keras, Hugging Face Transformers, Model Fine-tuning
NLP & LLMs	SentenceTransformers, RAG Systems, FAISS, Information Retrieval, Multilingual NLP
Computer Vision / Geospatial AI	OpenCV, Semantic Segmentation, Object Detection, Remote Sensing, GIS, QGIS, Google Earth Engine, Sentinel-2

LANGUAGES

Arabic Native French Fluent English Fluent